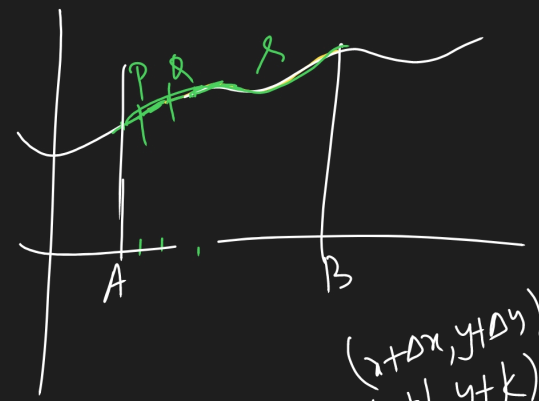
22/4/2026

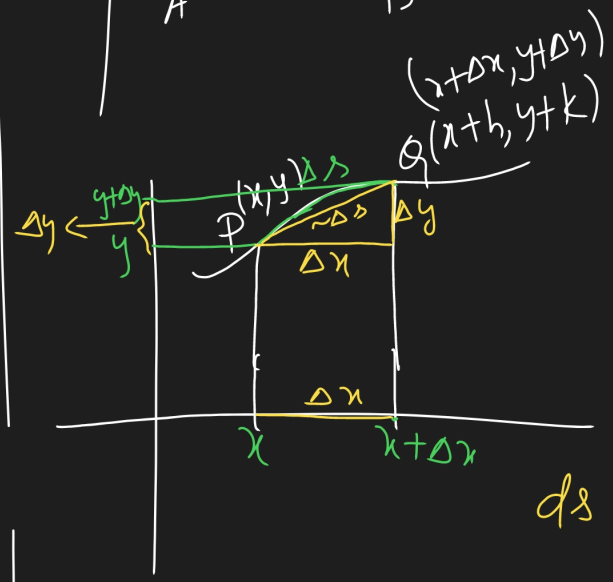
Rectification:

Here we find the lengths of given curves between any 2 points A & B

ie: 'Rectify' a given curve means finding the length of the curve

the conditions





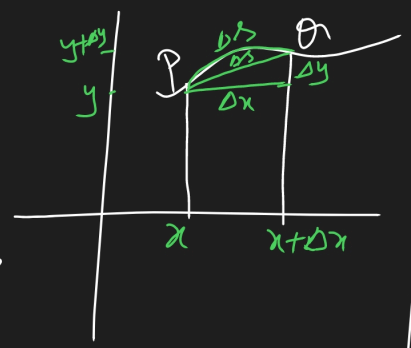
$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$$

$$\frac{(\Delta s)^2}{(\Delta x)^2} = 1 + \left(\frac{\Delta y}{\Delta x}\right)^2$$

$$ds \leftarrow \frac{\Delta s}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

$$\Rightarrow s = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

To find the length 's' of a given curve, we may first consider 2 neighbouring points $P(x, y)$ & $Q(x+\Delta x, y+\Delta y)$ which gives



$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$$

(by Pythagoras thm)

$$\Rightarrow \left(\frac{\Delta s}{\Delta x}\right)^2 = 1 + \left(\frac{\Delta y}{\Delta x}\right)^2$$

$$\Rightarrow \frac{\Delta s}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}$$

$$\Rightarrow s = \int_{A\, x_1}^{B\, x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dy \quad \text{— (1)}$$

<u>Similarly</u>:

$$s = \int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dx \quad \text{— (2)}$$

1. $\int \log x \, dx = x\log x - x$

$\boxed{X}$ 2. $\int \sqrt{a^2 + x^2} \, dx = \frac{x}{2}\left[\sqrt{a^2+x^2}\right] + \frac{a^2}{2}\left[\ln\sqrt{a^2+x^2} + x\right]$.

[Substitute $x = a\tan\theta$]

$$\int x^{3/2} dx = \frac{x^{5/2}}{5} \times 2$$

$$\int x^{-3/2} dx = \frac{x^{-1/2}}{-1} \times 2$$